Analiza orbita Orionida od 2007. do 2011. godine

ANTON PERKOV, GIMNAZIJA PULA, IV.RAZRED

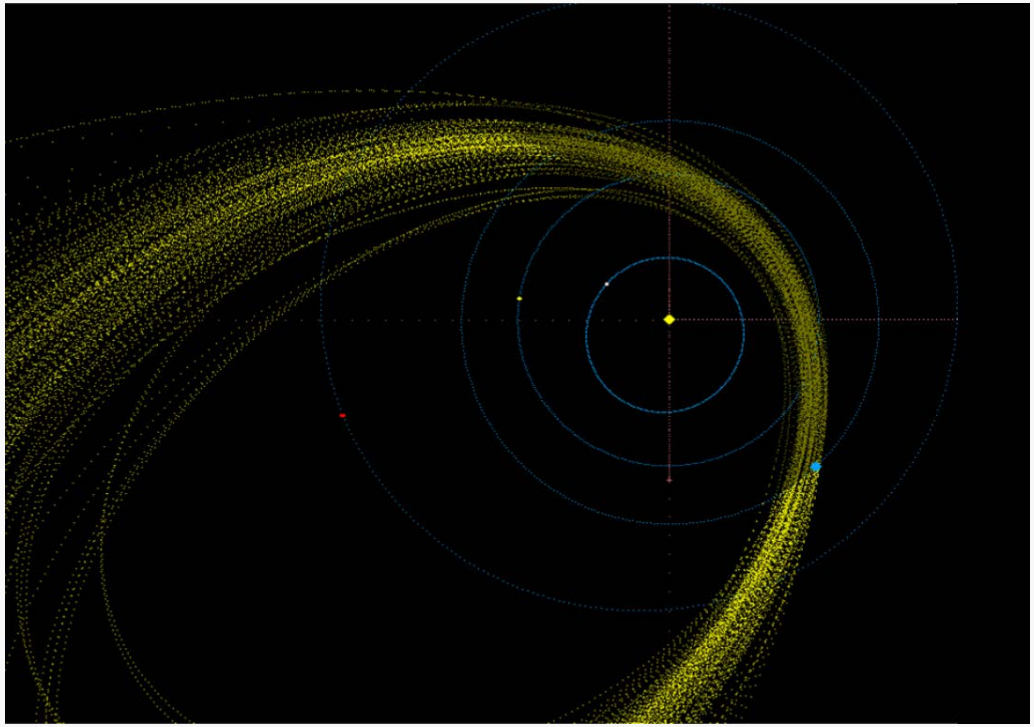
ASTRONOMSKO DRUŠTVO „ISTRA”, PULA

MENTOR: DAMIR ŠEGON, SSS

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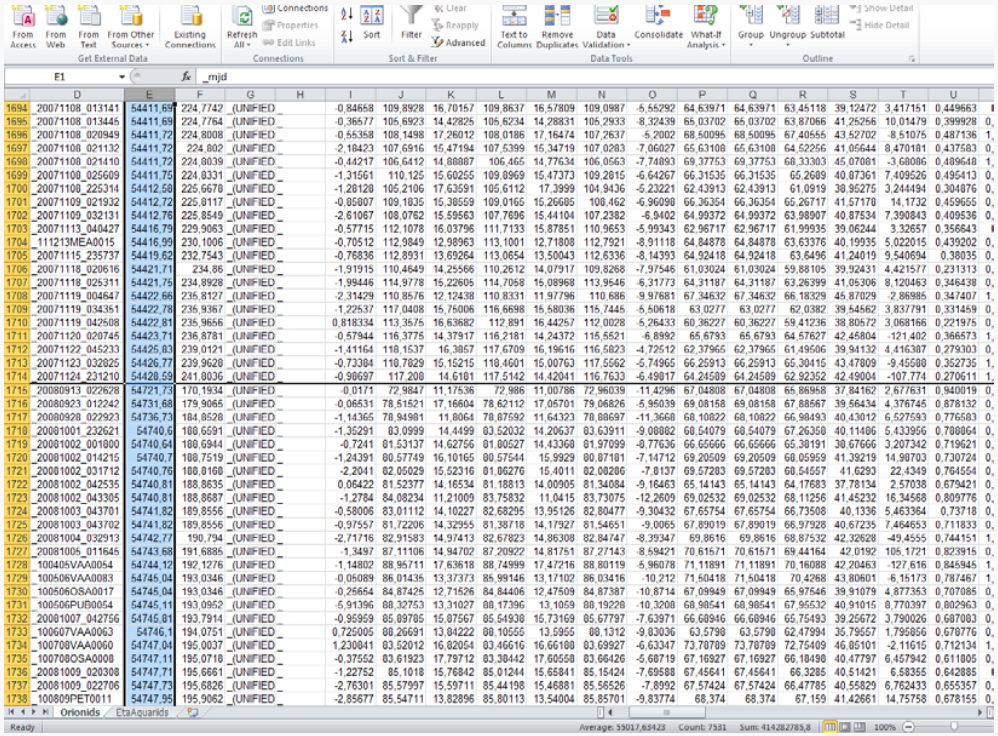
Uvod

- Porijeklo meteoroida
- Rezonantna struktura kometa 1P/Halley
- Vlakna meteoroida



Cilj

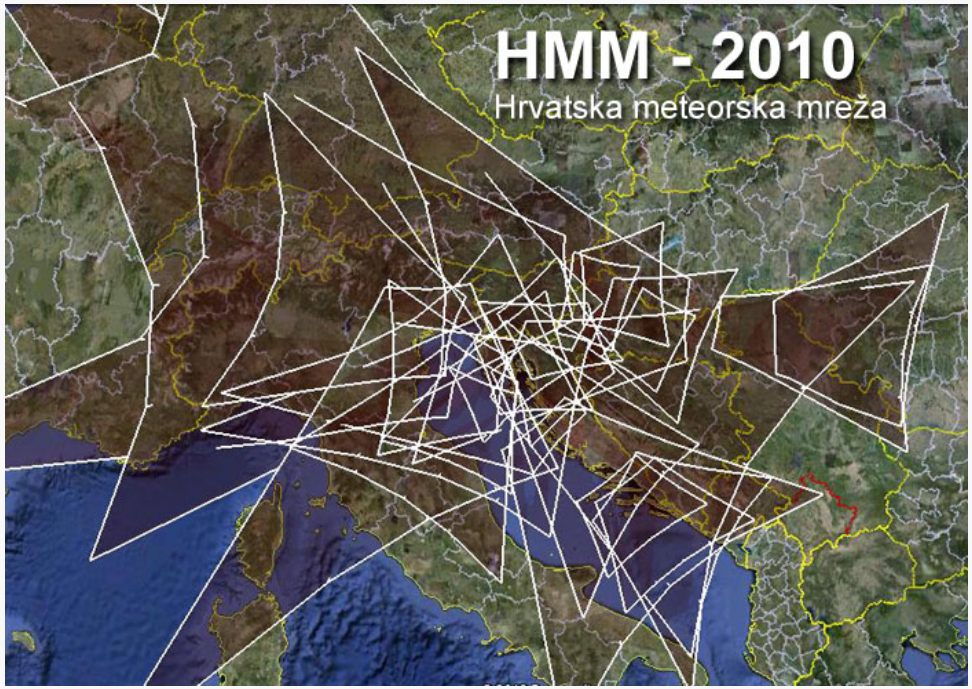
- Podaci o elementima orbite
- Utvrđivanje postojanja razlike tijekom godina



	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1694	_20071108_013141	54411,69	224,7742	_(UNIFIED_		-0,84658	109,8928	16,70157	109,8637	16,57809	109,0987	-5,55292	64,63971	64,63971	63,45118	39,12472	3,417151	0,449663
1695	_20071108_013445	54411,69	224,7764	_(UNIFIED_		-0,36577	105,6923	14,42825	105,6234	14,28831	105,2933	-8,32439	65,03702	65,03702	63,87066	41,25256	10,01479	0,399928
1696	_20071108_020949	54411,72	224,8008	_(UNIFIED_		-0,55358	108,1498	17,26012	108,0186	17,16474	107,2637	-5,2002	68,50095	68,50095	67,40555	43,52702	-8,51075	0,487136
1697	_20071108_021132	54411,72	224,802	_(UNIFIED_		-2,18423	107,6916	15,47194	107,5399	15,34719	107,0283	-7,06027	65,63108	65,63108	64,52256	41,05644	8,470181	0,437583
1698	_20071108_021410	54411,72	224,8039	_(UNIFIED_		-0,44217	106,6412	14,88887	106,465	14,77634	106,0563	-7,74893	69,37753	69,37753	68,33303	45,07081	-3,68086	0,489648
1699	_20071108_025609	54411,75	224,8331	_(UNIFIED_		-1,31561	110,125	15,60255	109,8969	15,47373	109,2815	-6,64267	66,31535	66,31535	65,2689	40,87361	7,409526	0,495413
1700	_20071108_225314	54412,58	225,6678	_(UNIFIED_		-1,28128	105,2106	17,63591	105,6112	17,3999	104,9436	-5,23221	62,43913	62,43913	61,0919	38,95275	3,244494	0,304876
1701	_20071109_021932	54412,72	225,8117	_(UNIFIED_		-0,85007	109,1835	15,38559	109,0165	15,29685	108,462	-6,96098	66,36354	66,36354	65,26717	41,57178	14,1732	0,459655
1702	_20071109_032131	54412,76	225,8549	_(UNIFIED_		-2,61067	108,0762	15,59563	107,7696	15,44104	107,2382	-6,9402	64,99372	64,99372	63,98907	40,87534	7,390843	0,409536
1703	_20071113_040427	54416,79	229,9063	_(UNIFIED_		-0,57715	112,1078	16,03796	111,7133	15,87851	110,9653	-5,99343	62,96717	62,96717	61,99935	39,06244	3,32657	0,356643
1704	_111213MEA0015	54416,99	230,1006	_(UNIFIED_		-0,70512	112,9849	12,98963	113,1001	12,71808	112,7921	-8,91118	64,84878	64,84878	63,63376	40,19935	5,022015	0,439202
1705	_20071115_235737	54419,62	232,7543	_(UNIFIED_		-0,76836	112,8931	13,69264	113,0654	13,50043	112,6336	-8,14393	64,92418	64,92418	63,6496	41,24019	9,540694	0,38035
1706	_20071118_020616	54421,71	234,86	_(UNIFIED_		-1,91915	110,4649	14,25566	110,2612	14,07917	109,8268	-7,97546	61,03024	61,03024	59,88105	39,92431	4,421577	0,231313
1707	_20071118_025311	54421,75	234,8928	_(UNIFIED_		-1,99446	114,9778	15,22605	114,7058	15,08968	113,9546	-6,31773	64,31187	64,31187	63,26399	41,05306	8,120463	0,346438
1708	_20071119_004647	54422,66	235,8127	_(UNIFIED_		-2,31429	110,8576	12,12438	110,8331	11,97796	110,686	-9,97681	67,34632	67,34632	66,18329	45,87029	-2,86985	0,347407
1709	_20071119_034351	54422,78	235,9367	_(UNIFIED_		-1,22537	117,0408	15,75006	116,6698	15,58036	115,7445	-5,50618	63,0277	63,0277	62,0382	39,54562	3,837791	0,331459
1710	_20071119_042508	54422,81	235,9656	_(UNIFIED_		0,818334	113,3575	16,63682	112,891	16,44257	112,0028	-5,26433	60,36227	60,36227	59,41236	38,80572	3,068166	0,221975
1711	_20071120_020745	54423,71	236,8781	_(UNIFIED_		-0,57944	116,3775	14,37917	116,2181	14,24372	115,5521	-6,8992	65,6793	65,6793	64,57627	42,45804	-121,402	0,366573
1712	_20071122_045233	54425,83	239,0121	_(UNIFIED_		-1,41164	118,1537	16,3857	117,6709	16,19616	116,5823	-4,72512	62,37965	62,37965	61,49506	39,94132	4,416387	0,279303
1713	_20071123_032825	54426,77	239,9628	_(UNIFIED_		-0,73384	118,7829	15,15215	118,4601	15,00763	117,5562	-5,74965	66,25913	66,25913	65,30415	43,47809	-9,45588	0,352735
1714	_20071124_231210	54428,59	241,8036	_(UNIFIED_		-0,98697	117,208	14,6181	117,5142	14,42041	116,7633	-6,49817	64,24589	64,24589	62,92352	42,49004	-107,774	0,270611
1715	_20080913_022628	54721,73	170,1934	_(UNIFIED_		-0,0171	72,9847	11,17536	72,986	11,00786	72,96039	-11,4296	67,04808	67,04808	65,86958	37,84162	2,677631	0,940019
1716	_20080923_012242	54731,68	179,9065	_(UNIFIED_		-0,06531	78,51521	17,16604	78,62112	17,05701	79,06826	-5,95039	69,08158	69,08158	67,88567	39,56434	4,376745	0,878132
1717	_20080926_022923	54736,73	184,8528	_(UNIFIED_		-1,14365	78,94981	11,8064	78,87592	11,64323	78,88697	-11,3668	68,10822	68,10822	66,98493	40,43012	6,527593	0,776583
1718	_20081001_232621	54740,6	188,6591	_(UNIFIED_		-1,35291	83,0999	14,4499	83,52032	14,20637	83,63911	-9,08802	68,54079	68,54079	67,26368	40,11486	5,433956	0,788064
1719	_20081002_001800	54740,64	188,6944	_(UNIFIED_		-0,7241	81,53137	14,62756	81,80527	14,43368	81,97099	-8,77636	66,65666	66,65666	65,38191	38,67666	3,207342	0,719621
1720	_20081002_014215	54740,7	188,7519	_(UNIFIED_		-1,24391	80,57749	16,10165	80,57544	15,9929	80,87181	-7,14712	69,20509	69,20509	68,05959	41,39219	14,98703	0,730724
1721	_20081002_031712	54740,76	188,8168	_(UNIFIED_		-2,2041	82,05029	15,52316	81,86276	15,4011	82,08286	-7,0137	69,57283	69,57283	68,54557	41,6293	22,4349	0,764554
1722	_20081002_042535	54740,81	188,8635	_(UNIFIED_		0,06422	81,52377	14,16534	81,18813	14,00905	81,34084	-9,16463	65,14143	65,14143	64,17683	37,78134	2,57038	0,679421
1723	_20081002_043305	54740,81	188,8687	_(UNIFIED_		-1,2784	84,08234	11,21009	83,75832	11,0415	83,73075	-12,2609	69,02532	69,02532	68,11256	41,45232	16,34568	0,809776
1724	_20081003_043701	54741,82	189,8556	_(UNIFIED_		-0,58006	83,01112	14,10227	82,68295	13,95126	82,80477	-9,30432	67,65754	67,65754	66,73508	40,1336	5,463364	0,73718
1725	_20081003_043702	54741,82	189,8556	_(UNIFIED_		-0,97557	81,72206	14,32955	81,38718	14,17927	81,54651	-9,0065	67,89019	67,89019	66,97928	40,67235	7,464653	0,711833
1726	_20081004_032913	54742,77	190,794	_(UNIFIED_		-2,71716	82,91583	14,97413	82,67823	14,86308	82,84747	-8,39347	69,8616	69,8616	68,87532	42,32628	-49,4555	0,744151
1727	_20081005_011646	54743,68	191,6885	_(UNIFIED_		-1,3497	87,11106	14,94702	87,20922	14,81751	87,27143	-8,59421	70,61571	70,61571	69,44164	42,0192	105,1721	0,823915
1728	100405VAA0054	54744,12	192,1276	_(UNIFIED_		-1,14802	88,95711	17,63618	88,74999	17,47216	88,80119	-5,96078	71,11891	71,11891	70,16088	42,20463	-127,616	0,845945
1729	100506VAA0083	54745,04	193,0346	_(UNIFIED_		-0,05089	86,01436	13,37373	85,99146	13,17102	86,03416	-10,212	71,50418	71,50418	70,4268	43,80601	-6,15173	0,787467
1730	100506OSA0017	54745,04	193,0346	_(UNIFIED_		-0,25654	84,87425	12,71526	84,84406	12,47509	84,87387	-10,8714	67,09949	67,09949	66,97546	39,91079	4,877353	0,707086
1731	100506PUI0054	54745,11	193,0952	_(UNIFIED_		-5,91396	88,32753	13,31027	88,17396	13,1059	88,19228	-10,3208	68,98541	68,98541	67,95532	40,91015	8,770397	0,802963
1732	_20081007_042756	54745,81	193,7914	_(UNIFIED_		-0,95959	85,89785	15,87567	85,54938	15,73169	85,67797	-7,63971	66,68946	66,68946	65,75493	39,25672	3,790026	0,687083
1733	100607VAA0063	54746,1	194,0751	_(UNIFIED_		0,725005	88,26691	13,84222	88,10555	13,5955	88,1312	-9,83036	63,5798	63,5798	62,47994	35,79557	1,796856	0,676776
1734	100708VAA0060	54747,04	195,0037	_(UNIFIED_		1,230841	83,52012	16,82054	83,46616	16,66188	83,69927	-6,63347	73,78789	73,78789	72,75409	46,85101	-2,11615	0,712134
1735	100708OSA0008	54747,11	195,0718	_(UNIFIED_		-0,37552	83,61923	17,79712	83,38442	17,60558	83,66426	-5,68719	67,16927	67,16927	66,18498	40,47797	6,457942	0,611805
1736	_20081009_020308	54747,71	195,6661	_(UNIFIED_		-1,22752	85,1018	15,76842	85,01244	15,65841	85,15424	-7,69588	67,45641	67,45641	66,3285	40,51421	6,58355	0,642885
1737	_20081009_022706	54747,73	195,6826	_(UNIFIED_		-2,76301	85,57997	15,59711	85,44198	15,46881	85,56526	-7,8992	67,57424	67,57424	66,47785	40,55829	6,762433	0,655357
1738	100809PET0011	54747,95	195,9062	_(UNIFIED_		-2,85677	85,54711	13,82896	85,80113	13,54004	85,85701	-9,83774	68,374	68,374	67,159	41,42661	14,75758	0,678155

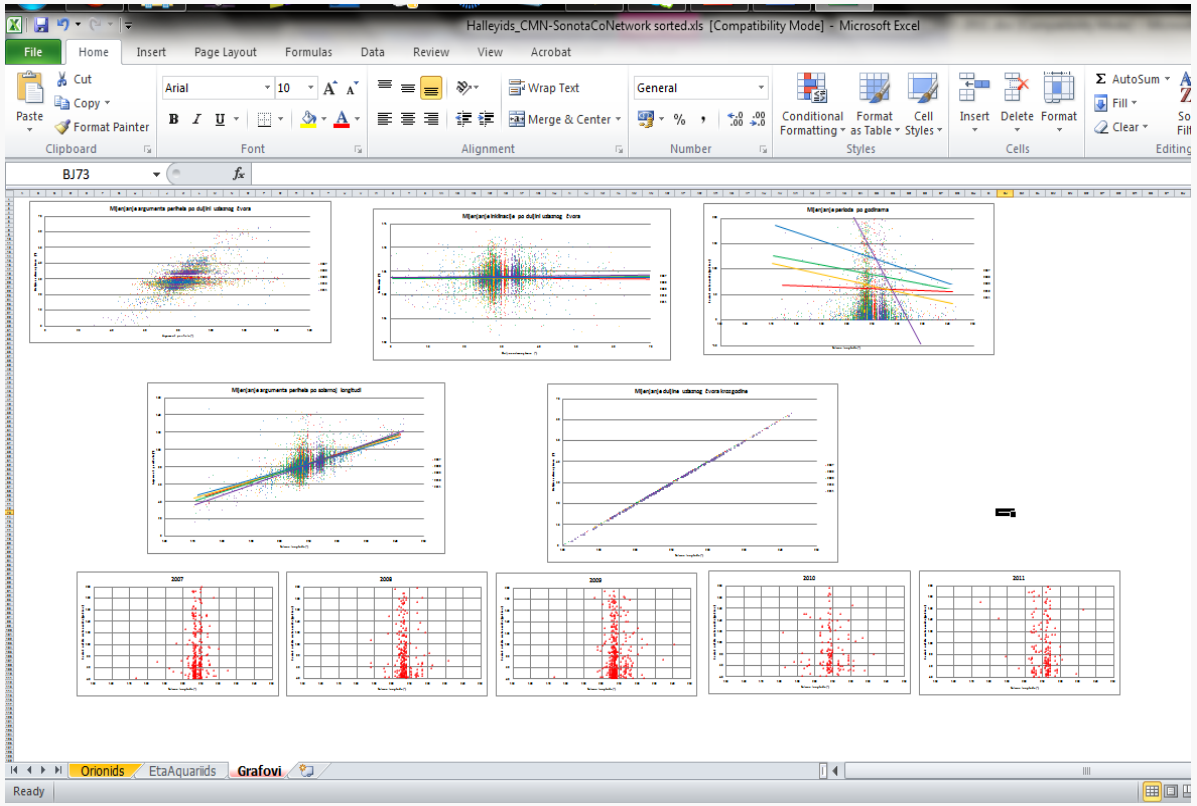
HMM

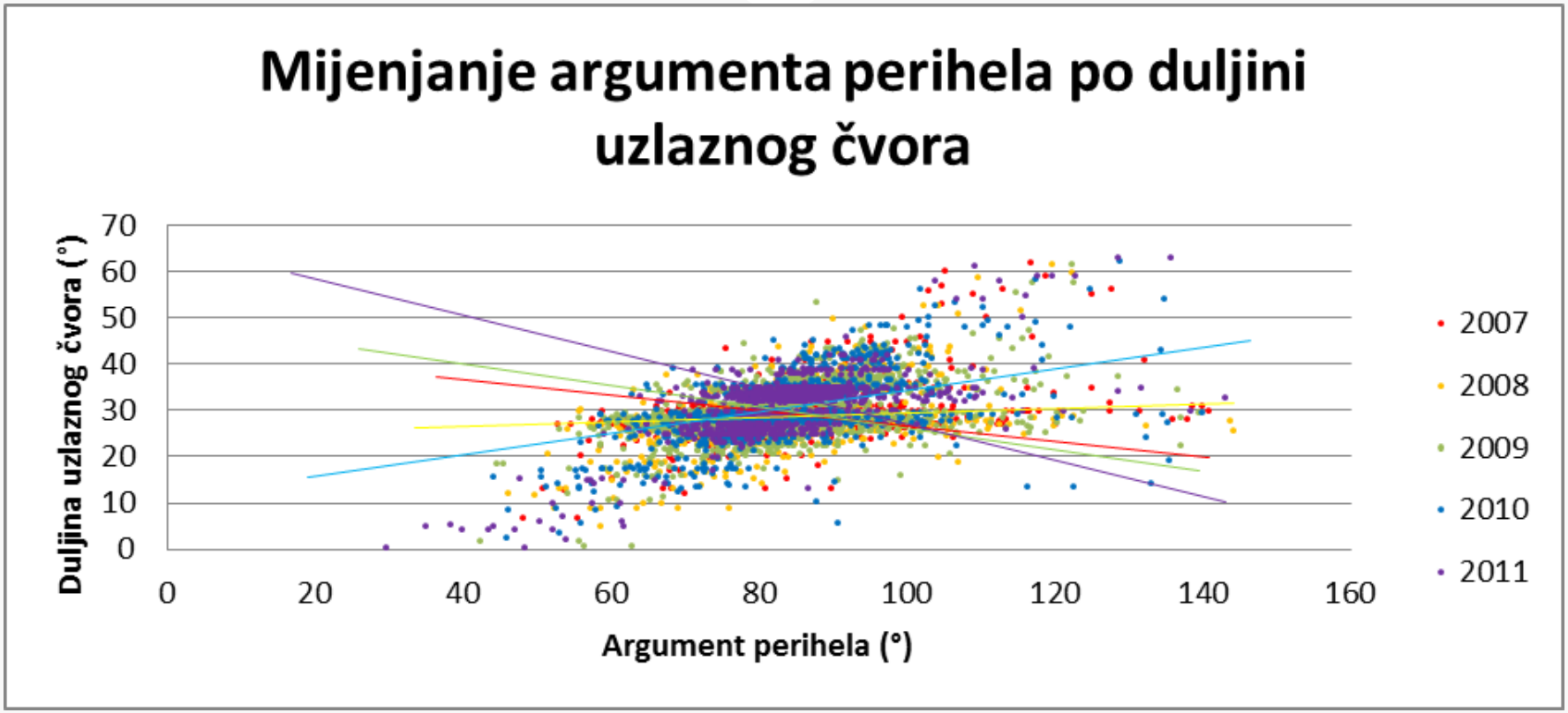
- Osnovana 2006. godine
- ~30 kamera u pogonu
- Svjetlosne pojave na nebu



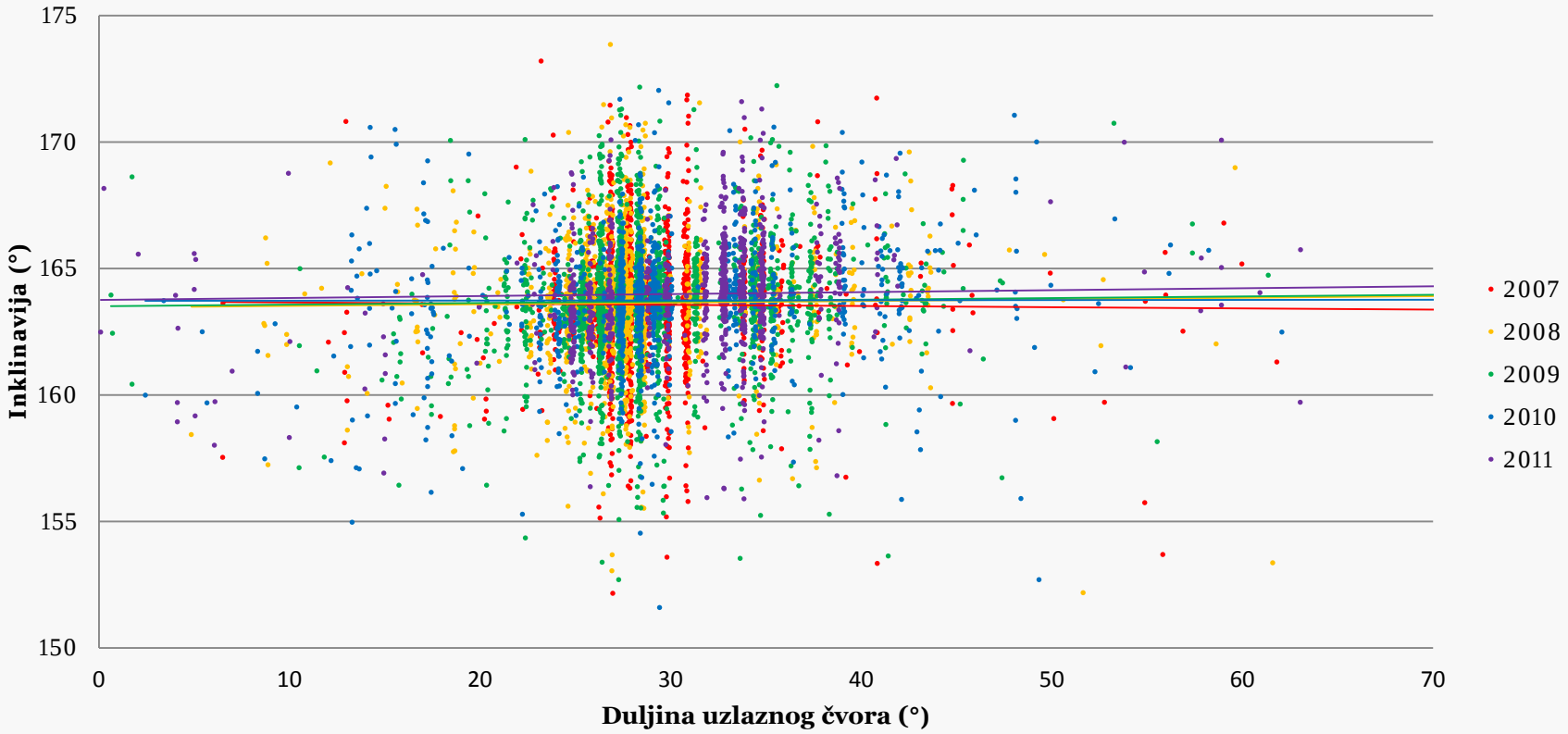
Obrada podataka

- Microsoft Excel
- Sortiranje
- Izrada grafova



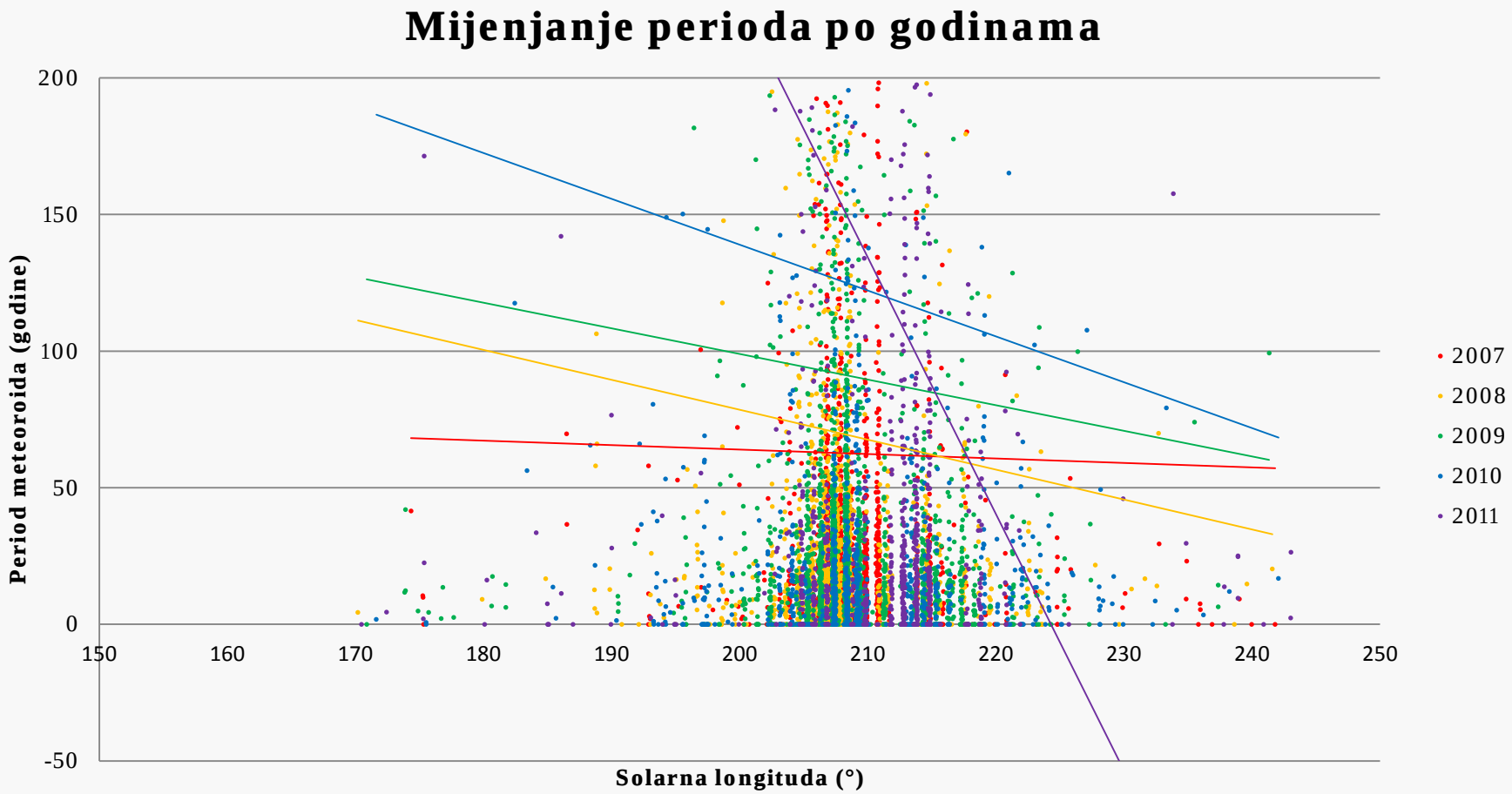

Mijenjanje argumenta perihela po duljini uzlaznog čvora
Duljina uzlaznog čvora (°)
70
60
50
40
30
20
10
0
0
20
40
60
80
100
120
140
160
Argument perihela (°)
2007
2008
2009
2010
2011

Mijenjanje inklinacije po duljini uzlaznog čvora



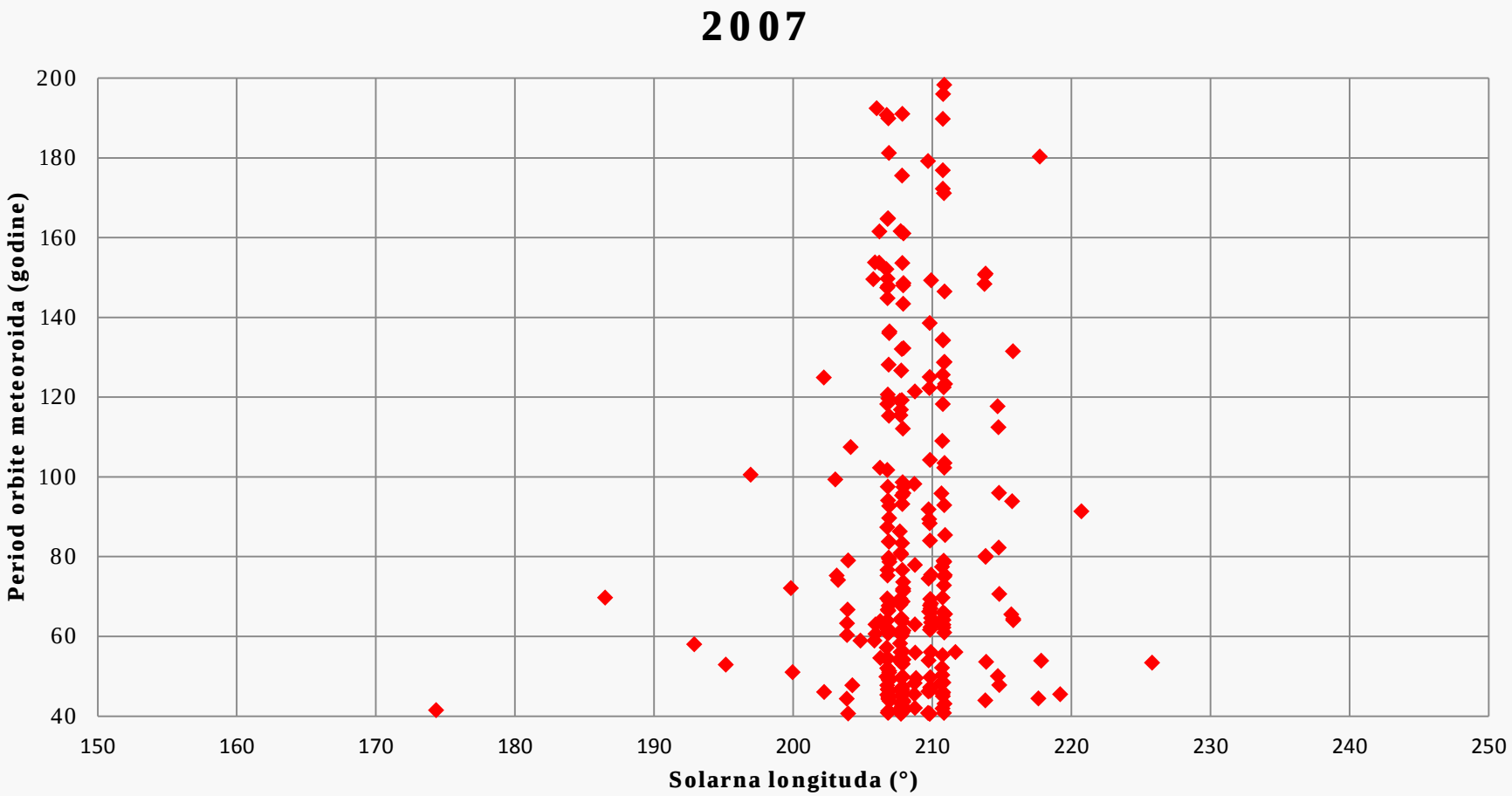
Anton Perkov: „Analiza orbita Orionida 2007.-2011. godine”, Gimnazija Pula

Problem velikih ulaznih brzina



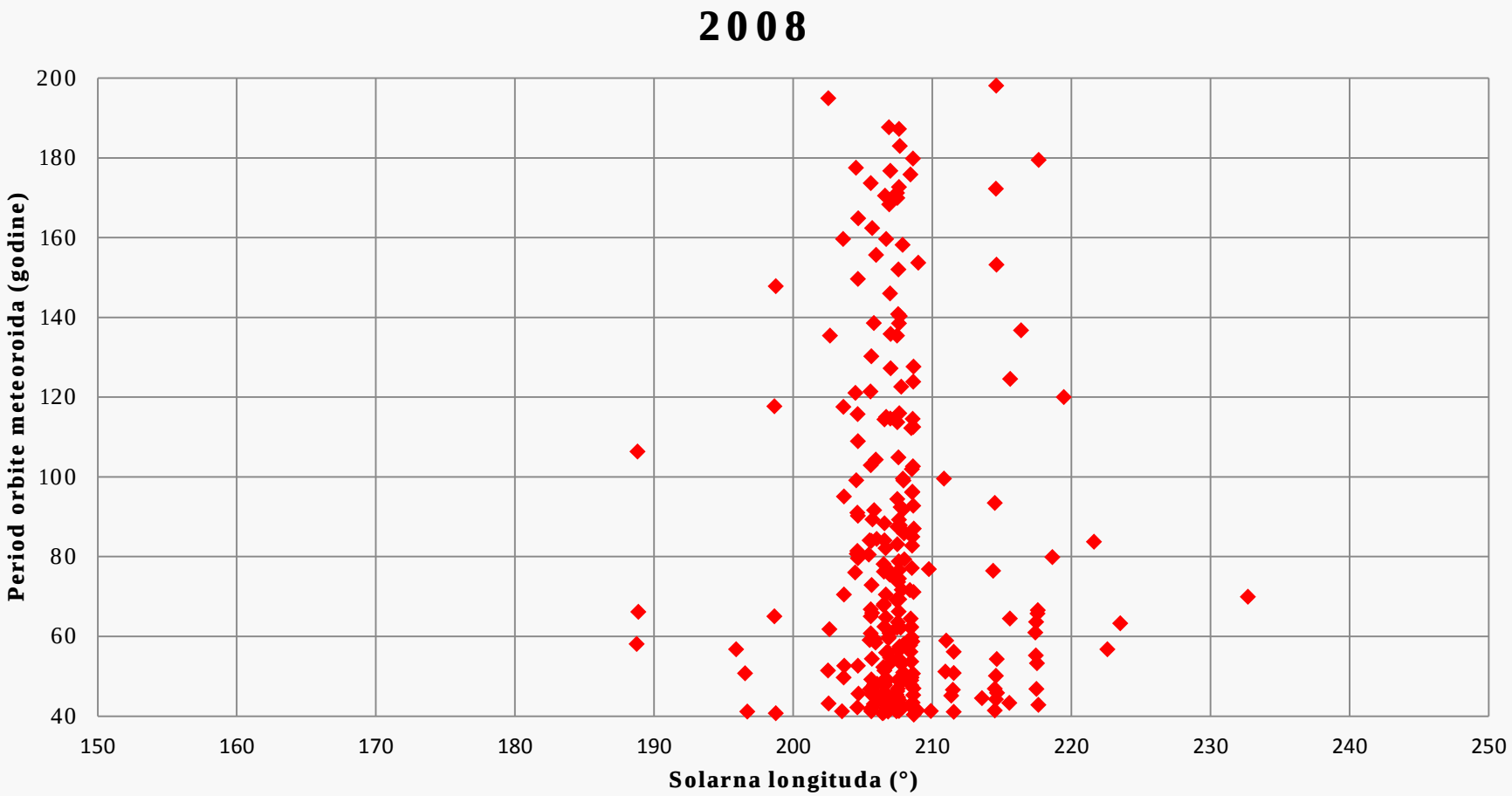
Anton Perkov: „Analiza orbita Orionida 2007.-2011. godine”, Gimnazija Pula

Pomicanje maksimuma aktivnosti tijekom godina



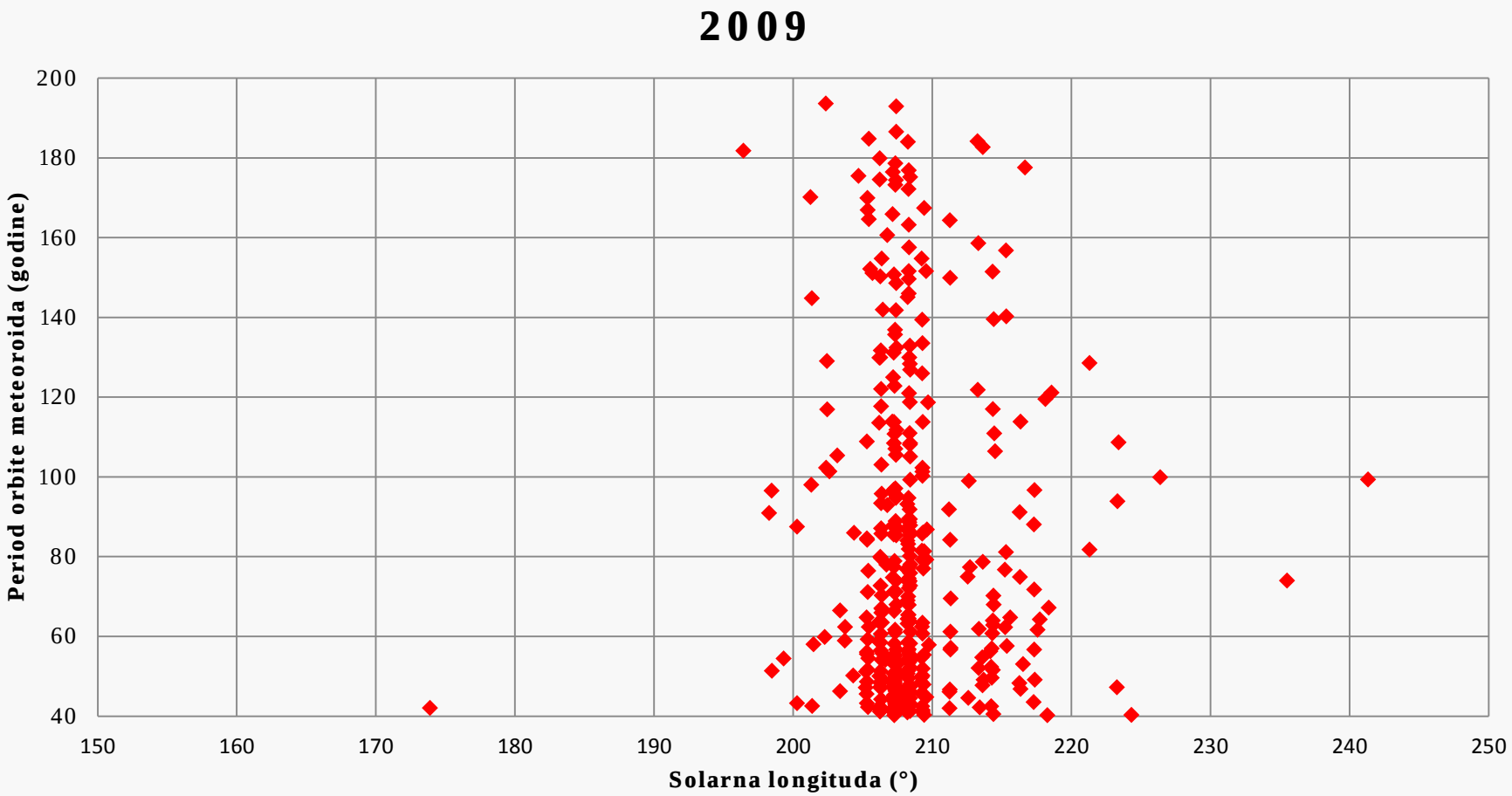
Anton Perkov: „Analiza orbita Orionida 2007.-2011. godine”, Gimnazija Pula

Pomicanje maksimuma aktivnosti tijekom godina



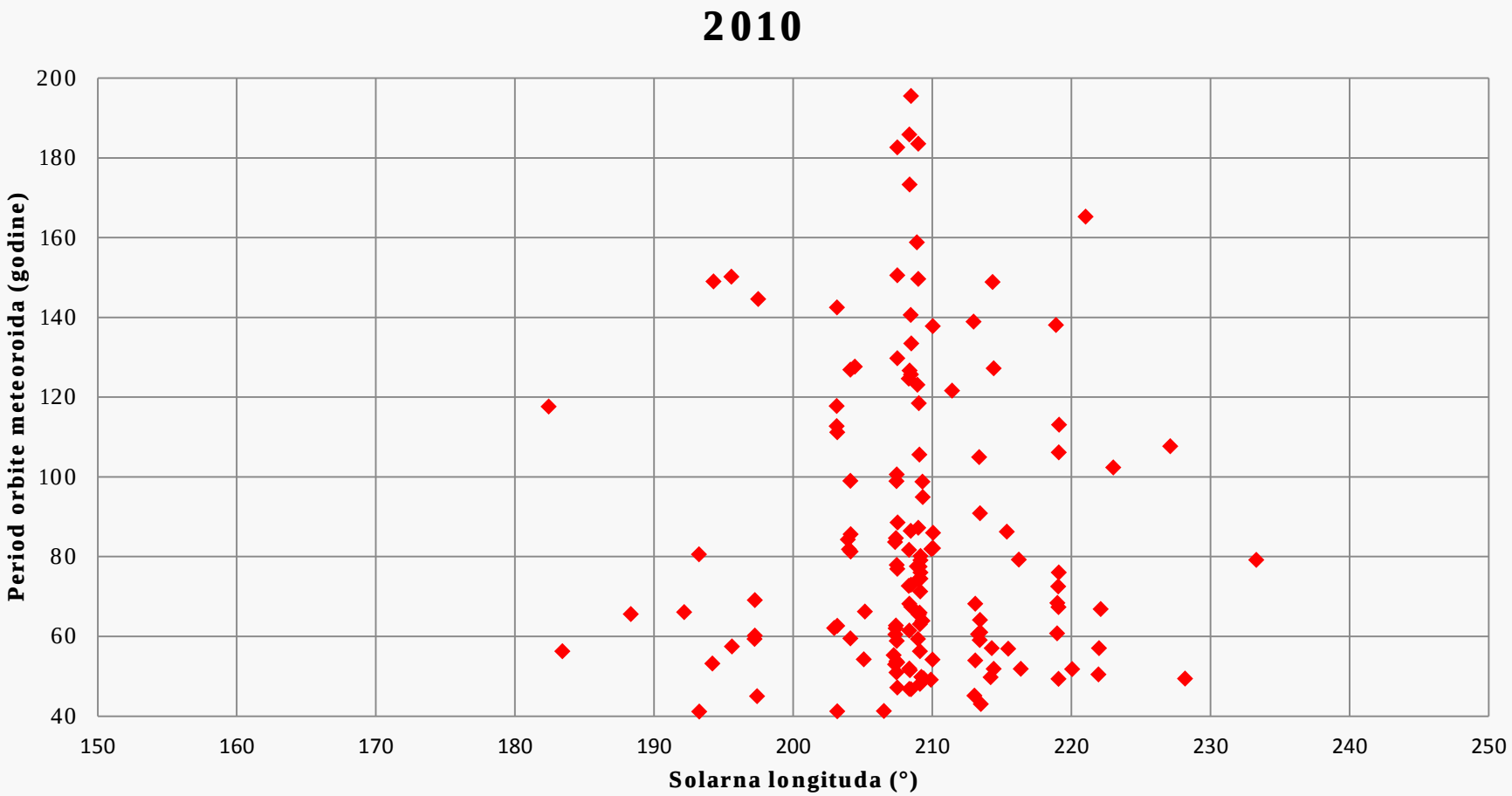
Anton Perkov: „Analiza orbita Orionida 2007.-2011. godine”, Gimnazija Pula

Pomicanje maksimuma aktivnosti tijekom godina



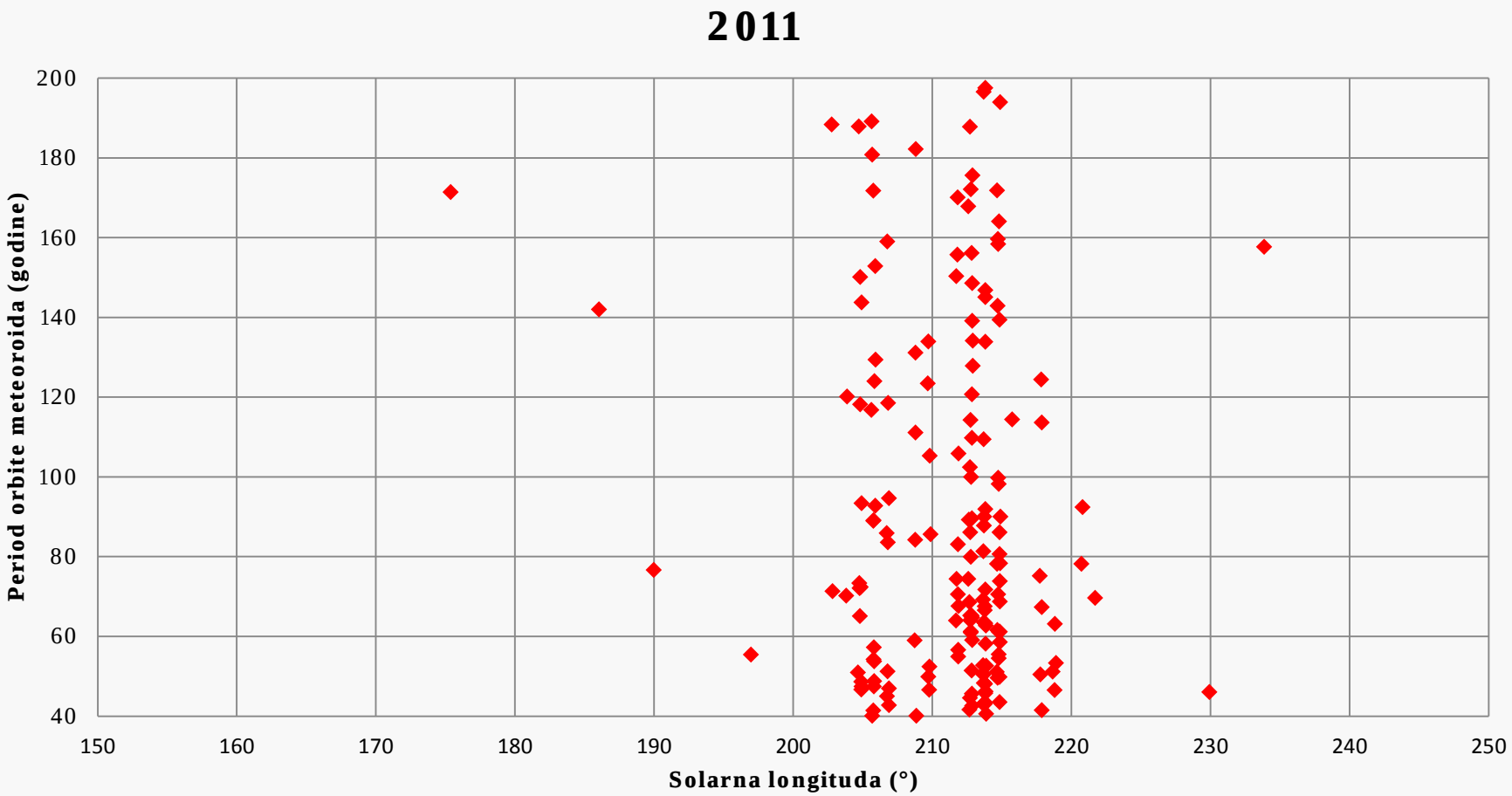
Anton Perkov: „Analiza orbita Orionida 2007.-2011. godine”, Gimnazija Pula

Pomicanje maksimuma aktivnosti tijekom godina



Anton Perkov: „Analiza orbita Orionida 2007.-2011. godine”, Gimnazija Pula

Pomicanje maksimuma aktivnosti tijekom godina



Anton Perkov: „Analiza orbita Orionida 2007.-2011. godine”, Gimnazija Pula

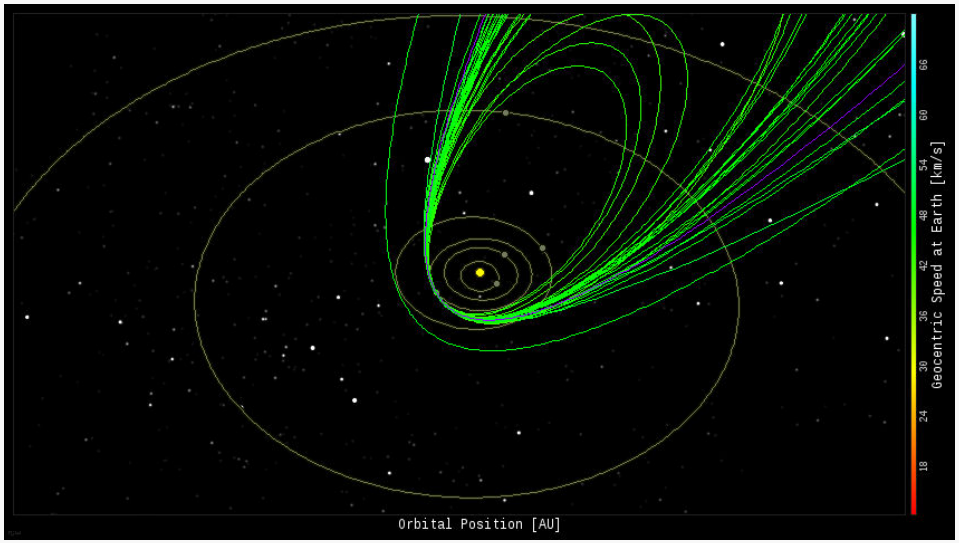
Zaključak



- Vlakna nerastegnuta po cijeloj orbiti - slaganje s predviđanjima iz literature
- Podaci svojstveni meteorskom potoku kao cjelini (inklinacija, argument perihela, ulazna brzina...) nepogodni za diferencijaciju vlakana

Daljnji planovi

- Simulacija kretanja Halleyeva kometa kroz vrijeme
- Mijenjanje perioda i orbite Halleya – rezonantna struktura
- Izrada prikaza današnje situacije




Hvala na pažnji!
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